

Links

LinkedIn:
linkedin.com/in/vinodneelakantam

GitHub:
github.com/vinodneelakantam

Skills

Build & Release Automation:
C/C++, Python, Groovy, Shell

Standards & Methodologies:
ASPICE SWE.5, ISO 21434, Agile, CI/CD/CT, TDD, SW/HW Debugging

Automotive Tools:
Development & Integration

DevOps for Embedded:
CMake, Jenkins (CloudBees), Docker/Kubernetes, Artifactory, Grafana/ElasticSearch

Infrastructure as Code:
Docker, Pipeline-as-Code (Groovy/Jenkinsfile/Github-Actions), Kyverno-Policy management

DevSecOps:
SAST/DAST, container & dependency scanning (Trivy/Snyk), secrets management, SBOM

Build Systems:
Yocto, Zephyr, AOSP, Custom (Bricks-Evo)

Cybersecurity:
OTA, Partial-Update Strategy

Security Features:
SecureAccess, SecureJTAG, Signing & Integrity of binaries

Interfaces:
UART, I2C, SPI, CAN, Ethernet

Vinod Kumar Neelakantam

EMBEDDED DEVOPS & BUILD ENGINEER | SOFTWARE INTEGRATION

Contact

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Personal Information

- **Nationality:** Indian (DE-Niederlassungserlaubnis)
- **Education:** Masters in Embedded Systems — Hyderabad, 2013
- **Languages:** English, German- A1
- **Driving License:** Class B

About Me

- 14+ years in automotive embedded software: ECU and ADAS integration, device drivers, board bring-up, system validation, platform release management.
- ASPICE SWE.5-aligned CI/CD architect: pipeline design, release ecosystems, quality gates, and traceability for distributed teams.
- Embedded build expertise: C/C++, CMake, Yocto, AOSP, Zephyr, Bricks-Evo; DevOps execution for multi-variant production platforms.
- System-level validation expert: HIL/SIL testing, smoke/regression/timing strategies, root cause analysis using Lauterbach, CANoe, ODIS.
- 7+ years automotive cybersecurity: TARA, ISO/SAE 21434, OTA/SOTA delivery, SecureAccess/SecureJTAG, binary signing and integrity controls.

Core Competencies

- Embedded Integration: ADAS, ECU platforms, multi-variant builds, dependency governance, cross-compilation.
- CI/CD & DevSecOps: Jenkins, CloudBees, Docker, Kubernetes, Artifactory, pipeline automation, SAST/SCA, SBOM, secrets management.
- Build Systems: CMake, Yocto, AOSP, Zephyr, Bricks-Evo, Infrastructure as Code, Dockerfiles, Kubernetes manifests, Pipeline-as-Code.
- Testing & Validation: HIL/SIL environments, VectorCAST, PRQA, smoke/regression/timing tests, validation via (CAN, Ethernet, SPI, I2C, UART).
- Automotive Security: TARA, ISO/SAE 21434, CSMS/SUMS (UNECE R155/R156), OTA/SOTA, SecureAccess, SecureJTAG, binary signing, integrity controls.
- Standards & Process: ASPICE SWE.5, audit preparation, release documentation, compliance frameworks.

Certifications

- ISTQB Certified Tester - Foundation Level (CTFL)

Current Role Focus

- Embedded Software Integration Engineer - Automotive/ADAS Platforms
- Build & Release Engineer - CI/CD, Embedded-DevOps
- Automotive Cybersecurity Engineer - Secure OTA/SOTA Delivery

Work Experience

Embedded Platform Integration

AUMOVIO (Continental) - Ulm

Sep 2021 - June 2026

- Implemented OTA-ready packaging workflows (PDX generation, signing, partial updates) in compliance with automotive cybersecurity requirements.
- Integrated and implemented security features (SecureAccess, SecureJTAG) with signing and integrity controls for binary artifacts across the platform.
- Developed Python-based tooling to automate build, integration, and release processes, and to analyze/optimize CI pipeline performance, improving efficiency, reliability, and traceability.
- Owned release lifecycle management, including dependency alignment, milestone tracking, and delivery execution.
- Set up and supported HIL/SIL environments and automated smoke testing for early validation during platform bring-up.
- Provided L2/L3 support for CI/CD infrastructure and build systems, and supported build issue debugging and system integration across complex embedded environments.
- Worked on automotive C/C++ build and integration workflows using Continental's Bricks-Evo custom build framework across ADAS platforms.
- Managed multi-variant software builds, handling dependencies, configuration, and cross-compilation for different ECU/product variants.
- Integrated and maintained CI/CD pipelines (Jenkins, CloudBees, Docker, Kubernetes, Artifactory), adopting Infrastructure as Code practices (Dockerfiles, Kubernetes manifests, Jenkins Pipeline-as-Code/Groovy) for version-controlled, reproducible, and auditable build and CI environments.
- Contributed to platform-level activities involving AOSP, Yocto, and Zephyr environments, including build setup and integration support.

Solution Integrator (ADAS, Surround View Systems)

AUMOVIO (Continental) - Ulm

Sep 2019 - Sep 2021

- Coordinated software releases with cross-functional teams (PMs, architects, testers) to meet project and customer milestones.
- Designed and maintained CI/CD pipelines (Jenkins, CMake, GitHub, Artifactory), codifying pipeline configuration as version-controlled Groovy scripts (Pipeline-as-Code) for repeatable, auditable builds.
- Defined and implemented ASPICE SWE.5 compliant integration processes, including build configuration, integration strategy, and release documentation.
- Developed and automated smoke, regression, and data-flow test strategies to ensure software quality and early defect detection.
- Set up and executed HIL/SIL validation environments during hardware bring-up and integration phases.
- Performed system-level debugging and root cause analysis using Lauterbach, CANoe, and ODIS tools.
- Conducted protocol validation across automotive interfaces (CAN, Ethernet, SPI, I2C, UART) ensuring reliable communication.
- Contributed to ADAS system integration (camera & surround view) for OEMs like Volkswagen and Audi.

Software Integrator (ADAS, Camera Mirror Replacement)

SMR Automotive Global - Stuttgart

Jul 2017 - Aug 2019

- Designed and optimized Jenkins-based CI/CD pipelines, improving build reliability and delivery efficiency for camera mirror systems.
- Defined ASPICE-compliant CI testing and integration strategies ensuring audit readiness and process consistency.
- Integrated testing and quality tools (VectorCAST, PRQA) into CI pipelines for automated validation and static analysis.
- Developed detailed integration and testing documentation including procedures, workflows, and technical concepts.
- Designed and executed comprehensive testing strategies (smoke, regression, timing, data flow) aligned with quality standards.
- Led ASPICE audit preparation and served as integration manager, achieving GREEN rating for customer deliveries; conducted GAP analysis and implemented process improvements to enhance development maturity and compliance.

Release Candidate Engineer (ARC Products)

Synopsys Inc - Hyderabad

Aug 2016 - Jul 2017

- Performed product verification and interoperability testing for Synopsys ARC release candidates across embedded hardware, software, operating systems, multimedia applications, and FPGA boards, safeguarding product suite integrity.

Software Integrator (Radars)

Infosys - Bengaluru

Mar 2014 - Aug 2016

- Collaborated with industry leaders APTIV and Caterpillar at their development centers to drive innovation and enhance efficiency in automotive software integration.
- Led the UDS fault management initiative and spearheaded UDS stack integration for Hyundai-Kia Motors' Short Range Side Radar (HKMC SRR3), ensuring optimal vehicle network performance.

Software Integrator (Engine Management Systems)

Robert Bosch Engineering - Coimbatore

Jun 2012 - Aug 2013

- Conducted thorough unit and integration testing, including Basic Functional Testing (BFT) and resource measurement, to ensure software integrity.
- Automated Smoke Testing procedures on the ETAS LAB Car, significantly increasing validation efficiency.